

Practice Question for MT-2

1. Multiply Ax by 26 using shift and addition instruction. Ax = 3
2. Write an assembly code to shift right and left a Dword (0 2345ABCD h) as Byte level 4times, considering no bit should be lost.
3. Write an assembly code to shift right and left a Dword (0 2345ABCD h) as Word level 2times, considering no bit should be lost.
4. The time stamp field of a file directory entry uses bits 0 through 4 for the seconds, bits 5 through 10 for the minutes, and bits 11 through 15 for the hours. (a) Write instructions that extract the seconds and copy the value to a byte variable named **BSeconds**. (b) Write instructions that extract the minutes and copy the value to a byte variable named **BMinutes**.

Consider 16 bits: DX = 0101100111011010

5. What will be the values of EDX:EAX after the following instructions execute?
mov edx,100h
mov eax,80000000h
sub eax,90000000h
sbb edx,0
6. What will be the contents of **DX**, **AX**, and the **Carry Flag (CF)** after executing the following instructions?
mov dx, 9A3Fh
mov ax, 7F80h
clc ; CF = 0
add ax, 90EAh ; step 1
adc dx, 1 ; step 2

stc ; CF = 1 (reset CF deliberately)
add ax, 8120h ; step 3
adc dx, 0FFFFh ; step 4 (note: adding 0xFFFF with carry)
7. What will be the contents of DX after the following instructions execute (STC sets the Carry flag)?
mov dx,5
stc ; set Carry flag
mov ax,10h
adc dx,ax
8. The following program is supposed to subtract val2 from val1. Find and correct all logic errors (CLC clears the Carry flag):

```

.data
val1 QWORD 20403004362047A1h
val2 QWORD 055210304A2630B2h
result QWORD 0

.code
mov cx,8          ; loop counter
mov esi,val1      ; set index to start
mov edi,val2
clc              ; clear Carry flag
top:
mov al,BYTE PTR[esi] ; get first number
sbb al,BYTE PTR[edi] ; subtract second
mov BYTE PTR[esi],al ; store the result
dec esi
dec edi
loop top

```

9. Write an assembly language program that calculates the area and perimeter of a rectangle using nested procedure. The main procedure should call CalculateRectangle. CalculateRectangle should call a nested procedure named FindArea that returns the area in EAX. CalculateRectangle should compute and display both area and perimeter.
Given:
length = 15, width = 10
10. Input a student's marks (0–100) and display grade:
 - A if ≥ 80
 - B if ≥ 60
 - C if ≥ 40
 - F otherwise
11. A smart traffic system reads a value representing **traffic load** (0–100).
If load > 70 → Light = RED
If load between 30–70 → Light = YELLOW
If load < 30 → Light = GREEN
Write an assembly program that displays which light should be ON.
12. Given an ASCII value in **Al**, classify it:
If it is a digit ('0'–'9'), jump to `digit`
If it is an uppercase letter ('A'–'Z'), jump to `upper`
If it is a lowercase letter ('a'–'z'), jump to `lower`
Otherwise, jump to `other`
13. Q-2: Given a number, check if **any bit among bit 2, 4, or 6** is 1. If yes → print "*At least one bit ON*", else "*All bits OFF*".

